MATH423: Introduction to Computational Topology

Mehmetcik Pamuk (mpamuk@metu.edu.tr)
Department of Mathematics, Middle East Technical University

2026-27 FALL

COURSE SYLLABUS

Course Description

Basic topology. Surfaces and their triangulations. Complexes. Homology. Persistence homology. Morse functions. Discrete Morse Functions. Applications.

Course Learning Outcomes

By the end of the semester, a successful student will be able to:

- Use efficient algorithms to compute the homology of a given complex.
- Compute persistence homology.
- Relate discrete Morse functions and persistence.
- Sketch persistence diagrams.

Suggested Books and Readings

While there is no single required textbook for this course, the following texts are highly recommended to supplement the lecture notes:

- H. Edelsbrunner and J. Harer, *Computational Topology: An Introduction*, American Mathematical Society, 2010.
- A. Zomorodian, *Topology for Computing*, Cambridge University Press, 2005.
- J. R. Munkres, *Elements of Algebraic Topology*, Addison-Wesley, 1984.
- L. C. Kinsey, *Topology of Surfaces*, Springer, 1993.

Week	Topic
1	Basic topology review: spaces, continuous maps, and homeomorphisms.
2	Surfaces and their triangulations: Euler characteristic and classification.
3	Abstract and geometric simplicial complexes, Vietoris-Rips and Čech complexes.
4	Cell complexes and their fundamental properties.
5	Homology: chain complexes, boundary operators, and cycles/boundaries.
6	Matrix reduction and efficient algorithms to compute homology.
7	Introduction to persistence homology: filtrations and persistent modules.
8	Computing persistent homology: Barcodes and sketching persistence diagrams.
9	Introduction to Morse functions: smooth manifolds and critical points.
10	Forman's Discrete Morse Theory: discrete Morse functions and gradients.
11	Discrete Morse complexes and topological simplification.
12	Relating discrete Morse functions and persistence homology.
13	Applications of computational topology in topological data analysis.
14	Further applications, review, and open problems.

Tentative Weekly Schedule

Course Assessment, Attendance, and Grading

The final grade for this course will be determined by two midterm exams and a final examination:

- **Midterm 1:** 30% (November 12, at 17:40)
- **Midterm 2:** 30% (December 17, at 17:40)
- **Final Exam:** 40% (Date: TBD)

Bonus Points: A total of 20 bonus points will be distributed across the two midterm exams. Consequently, the maximum possible raw score a student can achieve is 120 points. However, final evaluations and letter grades will be calculated out of a base score of 100 points.

Class Hours and Attendance Policy (NA Grade): This is a 3-hours-per-week course, and attendance will be taken at every lecture. If a student misses 10 lectures (lecture hours) during the semester, they will receive an **NA** grade from the course.

Letter Grade	Coefficient	Score Interval
AA	4.00	90 – 100
BA	3.50	85 – 89
BB	3.00	80 – 84
CB	2.50	75 – 79
CC	2.00	70 – 74
DC	1.50	65 – 69
DD	1.00	60 – 64
FD	0.50	50 – 59
FF	0.00	0 – 49
NA	0.00	–

Grading Scale: Letter grades will be assigned in accordance with the official METU catalog grading system. Based on a total of 100 points, the reference intervals are: